Probability & Statistics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in negative indices, evaluate $f(5)$.

Final Answer: 27. Substitute $x = 5$: $f(5) = 4(5) + 7 = 27$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 8$ in variables and indices, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for variables and indices consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in fractional indices (extended), evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for fractional indices (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 3$ in standard form, evaluate $f(2)$.

Final Answer: 7. Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for standard form consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A distribution of test scores has first quartile $Q_1 = 58$ and third quartile $Q_3 = 82$. Using the standard $1.5 \times \text{IQR}$ rule, what is the lowest score that would NOT be considered an outlier?

First compute the interquartile range (IQR): $\text{IQR} = Q_3 - Q_1 = 82 - 58 = 24$. Calculate the lower fence below which values are classified as low outliers: $\text{Lower Fence} = Q_1 - 1.5(\text{IQR}) = 58 - 1.5(24) = 58 - 36 = 22$. Any score at or above 22 is not a low outlier. The lowest non-outlier score is 22.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

Let $X \sim B(12, 0.05)$. We need $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$. $P(X = 0) = (0.95)^{12} \approx 0.5404$. $P(X = 1) = 12(0.05)(0.95)^{11} \approx 0.3413$. Thus, $P(X \geq 2) = 1 - (0.5404 + 0.3413) = 1 - 0.8817 = 0.118$ (3 s.f.).

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.

(a) $\binom{10}{6}(0.5)^{10} = 210 \times \frac{1}{1024} \approx 0.2051$. (b) $P(X \leq 2) = P(0) + P(1) + P(2) = \frac{1 + 10 + 45}{1024} = \frac{56}{1024} \approx 0.0547$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Statistics & Data Analysis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of μ and σ.

Standardising: $\Phi\left(\frac{65-\mu}{\sigma}\right) = 0.8413 \Rightarrow \frac{65-\mu}{\sigma} = 1.0$, so $65 - \mu = \sigma$. For the lower tail: $P(X < 45) = 0.0228 \Rightarrow \Phi\left(\frac{45-\mu}{\sigma}\right) = 0.0228 \Rightarrow \frac{45-\mu}{\sigma} = -2.0$, so $\mu - 45 = 2\sigma$. Adding the two equations: $65 - 45 = 3\sigma \Rightarrow 20 = 3\sigma \Rightarrow \sigma = 6.67$. Then $\mu = 65 - 6.67 = 58.33$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Mensuration

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

Multiply numerator and denominator by the conjugate of the denominator, $2 + \sqrt{5}$: $\frac{3+\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{(3+\sqrt{5})(2+\sqrt{5})}{2^2-(\sqrt{5})^2}$ Expanding the numerator: $3(2) + 3\sqrt{5} + 2\sqrt{5} + (\sqrt{5})^2 = 6 + 5\sqrt{5} + 5 = 11 + 5\sqrt{5}$. The denominator is $4 - 5 = -1$. Therefore: $\frac{11+5\sqrt{5}}{-1} = -11 - 5\sqrt{5}$ Thus $p = -11$ and $q = -5$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Geometry

In ΔPQR, ray QS bisects $\angle Q$ with S lying on side PR. If $PQ = 14$, $QR = 21$, and $PR = 25$, find the length of segment PS.

Step 1: Apply the Angle Bisector Theorem: The bisector divides side PR into segments proportional to the adjacent sides: $\frac{PS}{SR} = \frac{PQ}{QR} = \frac{14}{21} = \frac{2}{3}$ Step 2: Express SR in terms of PS: Since $PS + SR = 25$, we have $SR = 25 - PS$: $\frac{PS}{25-PS} = \frac{2}{3} \Rightarrow 3PS = 2(25 - PS) = 50 - 2PS$ Step 3: Solve for PS: $5PS = 50 \Rightarrow PS = 10$ (and $SR = 15$).

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

Find the equation of the line that passes through $(4, -1)$ and is perpendicular to the line $2x - 3y = 9$. Express your answer in slope-intercept form.

1. Find the slope of the given line by converting to slope-intercept form: $-3y = -2x + 9 \Rightarrow y = \frac{2}{3}x - 3$. The slope of the given line is $m_1 = \frac{2}{3}$. 2. The perpendicular slope is the negative reciprocal: $m_\perp = -\frac{3}{2}$. 3. Use point-slope form with $(x_1, y_1) = (4, -1)$: $y - (-1) = -\frac{3}{2}(x - 4) \Rightarrow y + 1 = -\frac{3}{2}x + 6 \Rightarrow y = -\frac{3}{2}x + 5$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

1. The maximum occurs at the vertex time $t = -\frac{b}{2a}$: $t = -\frac{64}{2(-16)} = \frac{64}{32} = 2$ seconds. Substitute $t = 2$ to get maximum height: $h(2) = -16(2^2) + 64(2) + 80 = -64 + 128 + 80 = 144$ feet. 2. Ground impact occurs when $h(t) = 0$: $-16t^2 + 64t + 80 = 0 \Rightarrow t^2 - 4t - 5 = 0 \Rightarrow (t - 5)(t + 1) = 0$. Since time must be non-negative, $t = 5$ seconds.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

Use the general formula $a_n = a_1 r^{n-1}$: $a_3 = a_1 r^2 = 18, \quad a_6 = a_1 r^5 = 486$. Divide a_6 by a_3: $\frac{a_1 r^5}{a_1 r^2} = r^3 = \frac{486}{18} = 27 \Rightarrow r = 3$. Solve for a_1: $a_1(3^2) = 18 \Rightarrow 9a_1 = 18 \Rightarrow a_1 = 2$. Compute S_5 using the geometric series sum formula: $S_5 = \frac{a_1(r^5 - 1)}{r - 1} = \frac{2(3^5 - 1)}{3 - 1} = \frac{2(243 - 1)}{2} = 242$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number

Solve the nonlinear system $y = x^2 - 3x + 2$ and $y = 2x - 4$ algebraically. State all intersection points (x, y).

Equate the expressions for y: $x^2 - 3x + 2 = 2x - 4 \Rightarrow x^2 - 5x + 6 = 0$. Factor the quadratic: $(x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$. Find corresponding y-values using $y = 2x - 4$:For $x = 2$: $y = 2(2) - 4 = 0$, yielding intersection point $(2, 0)$.For $x = 3$: $y = 2(3) - 4 = 2$, yielding intersection point $(3, 2)$.The system intersects at $(2, 0)$ and $(3, 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in multiplying and dividing with indices, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in index laws, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplying and dividing with indices consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for index laws consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a grouped frequency table, a class interval is $20 \le x < 35$ with a frequency of 30. What is the frequency density of this bar on a histogram?

a. 1.5

b. **2.0** ✓

c. 0.5

d. 15

Class width = $35 - 20 = 15$. Frequency density = $\frac{\text{Frequency}}{\text{Class width}} = \frac{30}{15} = 2.0$. [M1 for freq/width, A1 for 2.0]

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random without replacement. What is the probability that BOTH marbles are red?

a. $\frac{20}{56}$ ✓ **b.** $\frac{25}{64}$

c. $\frac{5}{8}$ **d.** $\frac{22}{56}$

P(1st Red) = $\frac{5}{8}$. P(2nd Red | 1st Red) = $\frac{4}{7}$. P(Both Red) = $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$. [M1 A1]

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two unbiased 6-sided dice are rolled simultaneously. What is the probability that the sum of the numbers shown is equal to 9?

a. $\frac{1}{9}$ ✓ **b.** $\frac{1}{6}$

c. $\frac{5}{36}$ **d.** $\frac{1}{12}$

Favourable pairs summing to 9: (3,6), (4,5), (5,4), (6,3) = 4 outcomes out of 36. Probability = $\frac{4}{36} = \frac{1}{9}$. [M1 A1]

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a test, the mean mark of 15 girls was 72, and the mean mark of 10 boys was 68. What was the combined mean mark of all 25 students?

a. **70.4** ✓ **b.** 70.0

c. 70.8 **d.** 69.6

Total marks = $(15 \times 72) + (10 \times 68) = 1080 + 680 = 1760$. Combined mean = $\frac{1760}{25} = 70.4$. [M1 A1]

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the interquartile range (IQR) of the following set of data: 4, 7, 9, 12, 15, 18, 22.

a. **11** ✓ **b.** 7

c. 18 **d.** 14

Lower quartile $Q_1 = 7$, Upper quartile $Q_3 = 18$. IQR = $Q_3 - Q_1 = 18 - 7 = 11$. [M1 A1]

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two independent events A and B have $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{7}$ **b.** $\frac{1}{2}$

c. $\frac{1}{4}$ ✓ **d.** $\frac{3}{4}$

For independent events, $P(A \cap B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The probability that it rains on any given day is 0.3. What is the probability that it rains on at least one of the next two days?

a. **0.51** ✓ **b.** 0.60

c. 0.09 **d.** 0.49

P(no rain both days) = $0.7 \times 0.7 = 0.49$. P(at least one rain) = $1 - 0.49 = 0.51$. [M1 A1]

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The numbers 3, 5, 7, 8, x, 12 have a mean of 8. What is the value of x?

a. **13** ✓ **b.** 11

c. 14 **d.** 9

Total sum = $6 \times 8 = 48$. Known sum = $3 + 5 + 7 + 8 + 12 = 35$. $x = 48 - 35 = 13$. [M1 A1]

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student draws a pie chart to represent favourite sports. If 18 out of 72 students chose Basketball, what is the sector angle representing Basketball?

a. **90°** ✓ **b.** 45°

c. 72° **d.** 120°

Angle = $\frac{18}{72} \times 360^\circ = \frac{1}{4} \times 360^\circ = 90^\circ$. [M1 A1]

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cumulative frequency diagram is plotted for 80 students. At what cumulative frequency value do you read the median score?

a. 20

b. **40** ✓

c. 60

d. 80

The median of 80 data values is located at the 50% position: $\frac{80}{2} = 40$. [B1]

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.7$, find $P(A \cap B)$.

a. **0.28** ✓

b. 1.10

c. 0.30

d. 0.55

For independent events, $P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.7 = 0.28$. [B1]

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Events A and B are mutually exclusive. If $P(A) = 0.35$ and $P(B) = 0.45$, find $P(A \cup B)$.

a. **0.80** ✓

b. 0.1575

c. 0.70

d. 0.10

For mutually exclusive events, $P(A \cup B) = P(A) + P(B) = 0.35 + 0.45 = 0.80$. [B1]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A box contains 4 red balls, 5 green balls, and 3 yellow balls. One ball is chosen at random. What is the probability that it is NOT green?

a. $\frac{7}{12}$ ✓

b. $\frac{5}{12}$

c. $\frac{1}{3}$

d. $\frac{3}{4}$

Total balls = 12. Non-green balls = 4 + 3 = 7. Probability = $\frac{7}{12}$. [B1]

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

On a cumulative frequency curve representing marks of 120 students, the median mark corresponds to which cumulative frequency?

a. **60** ✓

b. 30

c. 90

d. 100

Median corresponds to 50% of total frequency: $\frac{120}{2} = 60$. [B1]

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

In a survey of 20 people, 10 own a car. Of those, 5 also own a bicycle. Find the probability that a randomly chosen car owner also owns a bicycle.

a. $\frac{3}{7}$

b. $\frac{1}{7}$

c. $\frac{2}{7}$

d. $\frac{1}{2}$ ✓

$P(\text{bike} \mid \text{car}) = \frac{5}{10} = \frac{1}{2}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number

A spinner has probability 0.35 of landing on Gold. If the spinner is spun 200 times, what is the expected number of times it lands on Gold?

a. **70** ✓

b. 80

c. 100

d. 65

Expected frequency = $n \times p = 200 \times 0.35 = 70$. [B1]

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A bag has 3 red and 2 blue balls. Two balls are drawn without replacement. Find the probability both are red.

a. $\frac{1}{10}$

b. $\frac{3}{10}$ ✓

c. $\frac{1}{2}$

d. $\frac{4}{5}$

$P(\text{RR}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Algebra & Graphs

The probability that it rains tomorrow is $\frac{4}{5}$. Find the probability that it does NOT rain.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{5}$ ✓

d. $\frac{2}{5}$

$P(\text{not rain}) = 1 - \frac{4}{5} = \frac{1}{5}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Coordinate Geometry

From a cumulative frequency graph, the lower quartile Q_1 is 28 and the upper quartile Q_3 is 54. Calculate the interquartile range (IQR).

a. **26** ✓

b. 28

c. 41

d. 82

Interquartile range = $Q_3 - Q_1 = 54 - 28 = 26$. [B1]

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

The frequency table shows ages of 50 museum visitors. Calculate (a) mean age, (b) modal class.

a. Mean ≈ 39, Modal class 40–59

b. **Mean ≈ 42, Modal class 40–59** ✓

c. Mean ≈ 39, Modal class 20–39

d. Mean ≈ 42, Modal class 20–39

Mean ≈ 42. Modal class: 40–59 (frequency 18).

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry

A data set has values and frequencies: $1(\times 4), 2(\times 3), 3(\times 2), 4(\times 1)$. Find the mean.

a. 5

b. **2** ✓

c. 11

d. 10

Mean $= \frac{\sum fx}{\sum f} = \frac{20}{10} = 2$.

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